

|                      | Quadrant | ER model                         | Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Reality              | Q1       | Event (E)                        | Eve logs on with her password (2024-10-04T16:00BST) and conducts a web search for 'how to blow up the moon' (2024-10-04T16:10BST)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | Q1       | Environment/<br>system<br>change | This causes many changes on the system. For example:<br>- SAM file last login updates;<br>- Security Event log updates;<br>- UserAssistKey update for Chrome run,<br>- Chrome history database;<br>- Chrome cache entries<br>...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                      | Q2       | Subsequent<br>events             | Eve logs on for several subsequent days before the machine is seized:<br>The SAM file last login is suppressed through overwriting;<br>- Event logs are added;<br>- Chrome history is added;<br>- Chrome cache entries are added.<br>...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                      | Q2       | Extractable<br>artifacts         | This results in the following being available:<br>- \$MFT (NTFS file system parsing)<br>- SAM (registry parsing)<br>- Security.evtx (event log parsing)<br>- Chrome history (history SQLite database parsing)<br>- Chrome cache (Chrome cache parsing)<br>...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Event Reconstruction | Q2/3     | Timeline<br>generation           | Here we have the data that results after the subsequent events are observed. This is forensic tool and technique specific as to what can be extracted. Data was acquired using X-Ways, and processed using plaso.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                      | Q3       | Timeline                         | The timeline generated is dependent on the tools/techniques used.<br>After parsing the artifacts, timestamps can be extracted from the parsed data, normalised and sorted.<br>Examples are (for plaso):<br>- File creation time (FILE, File stat)<br>- Event log entries (EVT, WinEVTX)<br>- Registry key last write times (REG, Registry Key)<br>- Chrome history events (WEBHISTORY, Chrome History)<br>- Chrome cache events (WEBHISTORY, Chrome Cache)                                                                                                                                                                                                                                                                                                                                    |
|                      | Q3/4     | Timeline<br>analysis             | Examination of the timeline entries can be done in a variety of ways. If the time of interest is known the timeline can be filtered. If it is not then a keyword search for 'moon' etc. may result in a hit in the Chrome history entries.<br>We observe:<br>- Event log entry login entry (ID4624) for Eve account at 15:00:25 (UTC)<br>- 5 mins later (15:10:02.979138 UTC) a Google search (Chrome history) recorded for how to blow up the moon recovered from Google history.                                                                                                                                                                                                                                                                                                            |
|                      | Q4       | Hypothesis                       | Several initial hypotheses can be generated at this time (which can vary with examiner knowledge and bias):<br><b>H1</b> Eve logged on at 15:00:25 with her password and searched for how to blow up the moon (15:10:02).<br>Considering alternative hypothesis:<br><b>H2</b> Someone else logged on with her password and searched for how to blow up the moon.<br><b>H3</b> Eve logged on and an automated process resulted in the chrome history entry                                                                                                                                                                                                                                                                                                                                     |
|                      | Q4       | Hypothesis<br>testing            | Considering the history, checking for additional timeline entries to support the Chrome search being 'fingers at keyboard' rather than an automated process.<br>In terms of timelines:<br>- Consider the history entry in more detail and check if 'transition' is 'redirect', 'bookmark', 'typed' etc.:<br>plaso indicates transition type is 'GENERATED' (TODO:manually check later if necessary).<br>- Consider Chrome cache contents at this time:<br>This results in incremental entries such as:<br>"www.google.com/complete...q=h",<br>" www.google.com/complete...q=ho",<br>" www.google.com/complete...q=how",<br>...<br>"www.google.com/complete...q=how+to+blow+up+the+moon".<br>More broadly with an additional non-timeline based toolset:<br>- image can be scanned for malware |
|                      | Q4       | Hypothesis                       | <b>H3</b> deemed less likely than <b>H1</b> or <b>H2</b> as no malware has been found, the Chrome history entry has flag ('GENERATED' from plaso). Chrome cache has incremental 'complete..q=' entries. <i>Follow up</i> :<br>Additional knowledge on Chrome cache entries (Google search API) is needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                      | Q4       | Hypothesis<br>testing            | Experiment in VM to slowly type a search, and consider the resulting cache entries. Experiment results are consistent with the data observed in Eve's disk image.<br>No other hypotheses for this data have been generated so current inferred events involve someone typing this query.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                      | Q4       | Hypothesis                       | <b>H1</b> Eve logged on at 15:00:25 with her password and then searched for how to blow up the moon at 15:10:02;<br><b>H2</b> Someone else logged on with her password and searched for how to blow up the moon at 15:10:02<br><b>H4</b> Eve logged on at 15:00:25 with her password, someone else searched on her machine at 15:10:02                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| Q4 | Hypothesis testing   | <p>Examine event log entry at 15:00:25<br/>         – is it local login or remote...<br/>         – research into event log codes...<br/>         – was local login (Type=2)</p> <p>Consider additional sources of evidence.<br/>         ... access card log shows her swiping into the building at 14:51, and the office at 14:57.</p> <p>Two other people in the room from those logs, Alice and Bob.</p> <p>Outside of timeline actions: Q is Eve's password secure?<br/>         – attempt hashcat break, and ask her what it is. -&gt; Unable to break password with lists available. Eve uncooperative.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Q4 | Hypothesis           | <p><b>H1</b> Eve logged on at 15:00:25 with her password and searched for how to blow up the moon<br/> <b>H2a</b> Alice logged on with Eve's password and searched for how to blow up the moon<br/> <b>H2b</b> Bob logged on with Eve's password and searched for how to blow up the moon<br/> <b>H4:</b> Eve logged on at 15:00:25 with her password, and either Alice or Bob searched for how to blow up the moon on her machine at 15:10:25.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Q4 | Hypothesis testing   | <p>Examination of machine timeline around search...</p> <p>Between login (15:00:25) and Google search (15:10:02), a directory 'meeting notes' has a created time of 15:08:25 and a file called 'Meeting Notes 3rd October.rtf' (and USN journal entries) has created time of 15:08:52. Outside of the timeline, the document content is a summary of a meeting that took place the day before (and supported by Windows shortcut files , 'Recent' registry keys, and automatic destinations at similar time).</p> <p>Contains information that Eve would know from attending the meeting. The document content notes that Alice and Bob were absent from the meeting. This is supporting evidence for <b>H1</b>.</p> <p>Examination of Alice's machine at the time shows user/file activity on her machine at the time of the search, and examination of Bob's machine at the time shows user/file activity on his machine at the time of the search. So there is contradictory evidence for <b>H2a</b> and <b>H2b</b>.</p> |
| Q4 | Hypothesis           | <p><b>H1</b> Eve logged on at 15:00:25 with her password and searched for how to blow up the moon at 15:10:02;<br/>         Reviewing the potential issues at each stage of the event generation/reconstruction cycle, alternate hypotheses that should be considered (from <i>Q1-Environment</i>):<br/> <b>H4</b> – Clock was not correct meaning interpretation and correlation with other activities was incorrect/irrelevant.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Q4 | Hypothesis testing   | Examination shows time synchronisation was enabled. Sync event occurred. Searching for time anomalies resulted in no results so there is no evidence for <b>H4</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|    | Hypothesis           | <b>H1</b> Eve logged on at 15:00:25 with her password and searched for how to blow up the moon at 15:10:02.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Q4 | Inferred Events (E') | <b>H1</b> Eve logged on at 15:00:25 with her password and searched for how to blow up the moon at 15:10:02.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |